The Eventual-Increment Spectrum of Cloitre's Recurrence Is Not Surjective

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Source, certificate, and verification code are in the repository listed in the references; the exact source revision for this draft is recorded there under `manuscript/arxiv/README.md`.
The universal stabilization conjecture remains open, and this draft has not yet received external specialist review.

Abstract

For a positive integer m, define

$$b_1 = m, \qquad b_{n+1} = b_n + (b_n \bmod n),$$

where $b_n \bmod n$ denotes the least nonnegative remainder. It is conjectured that every orbit eventually has constant first difference, but this remains open. We study instead the set of positive integers that occur as an eventual constant increment for at least one stabilizing orbit. We prove the effective bound

$$m < (c+3)(3c+5)$$

whenever the orbit starting from m stabilizes with eventual increment c. Thus the attainability of any fixed increment is a finite question. Combining this theorem with a complete, independently regenerated arbitrary-precision certificate for every start $1 \leq m < 260$, we prove that the attainable eventual-increment spectrum is not all of $\mathbb{Z}_{>0}$: the two smallest omitted values are 5 and 7. The conclusion is unconditional and does not assume universal stabilization.

Keywords. modular recurrence, integer dynamics, eventual periodicity, finite certificate, OEIS A073117, OEIS A117846.

1 Introduction

For each $m \in \mathbb{Z}_{>0}$, consider the integer recurrence

$$b_1 = m, \qquad b_{n+1} = b_n + (b_n \bmod n). \tag{1}$$

The recurrence appears in OEIS A073117 and A117846 and was discussed in MathOverflow question 191518; see [1, 2, 3]. Cloitre's stabilization conjecture asks whether every positive starting value m produces an orbit whose increments $b_{n+1} - b_n$ are eventually constant.

The conjecture itself is not resolved here. Our subject is the smaller and logically independent question of which constants can occur when stabilization does happen.

Definition 1.1 (Attainable eventual-increment spectrum)**.** Let $\mathcal{E}$ be the set of positive integers c for which there exist $m \geq 1$ and $t \geq 1$ such that

$$b_{n+1} - b_n = c \qquad \text{for all } n \geq t.$$

Abercrombie asked in OEIS A117846 whether these values include every positive integer [2]. Our main result answers this negatively.

Theorem 1.2 (Main theorem)**.** *The attainable spectrum $\mathcal{E}$ is not equal to $\mathbb{Z}_{>0}$. More precisely,*

$$1, 2, 3, 4, 6 \in \mathcal{E}, \qquad 5, 7 \notin \mathcal{E}.$$

Consequently, 5 *and* 7 *are the two smallest omitted eventual increments.*

The proof has two distinct components. The mathematical component is an effective bound: if a start m stabilizes with increment c, then $m < (c+3)(3c+5)$. The computational component is a complete exact certificate for all $1 \leq m < 260$. The certificate is sufficient because the theorem reduces the cases $c = 5$ and $c = 7$ to $m < 160$ and $m < 260$, respectively. No assumption is made about starts outside these finite intervals or about universal stabilization.

2 Basic coordinates and absorption

Write

$$b_n = q_n n + r_n, \qquad 0 \leq r_n < n, \tag{2}$$

so that $q_n = \lfloor b_n/n \rfloor$ and $r_n = b_n \bmod n$. Since $b_1 \bmod 1 = 0$, we have $b_2 = m$, so index 2 is the natural starting index for the quotient-remainder dynamics.

From (1) and (2),

$$b_{n+1} = q_n n + 2r_n = q_n(n+1) + (2r_n - q_n).$$

Therefore

$$q_{n+1} = q_n + \left\lfloor \frac{2r_n - q_n}{n+1} \right\rfloor, \qquad r_{n+1} = (2r_n - q_n) \bmod (n+1). \tag{3}$$

Theorem 2.1 (Absorption criterion)**.** *Fix an index t. The following statements are equivalent:*

(i) $b_{n+1} - b_n = c$ *for every* $n \geq t$*;*

(ii) $b_t = c(t+1)$ *with* $0 \leq c < t$*;*

(iii) $q_t = r_t = c$*.*

Proof. If $q_t = r_t = c$, then $b_t = ct + c = c(t+1)$ and $c < t$. Conversely, suppose $b_n = c(n+1)$ and $c < n$. Then $b_n \bmod n = c$, so

$$b_{n+1} = c(n+1) + c = c(n+2),$$

and induction gives constant increment c forever.

Finally, suppose the increment is c for all $n \geq t$. Then $r_n = c$ and

$$b_n = b_t + c(n-t).$$

Since $b_n = q_n n + c$, we obtain

$$q_n = c + \frac{b_t - ct - c}{n}.$$

The left side is an integer for every sufficiently large n, while the fraction tends to zero. Its numerator must vanish. Thus $b_t = c(t+1)$, and $c = r_t < t$. □

The set $q_n = r_n$ is therefore absorbing. It is useful to introduce the signed coordinate

$$e_n = r_n - q_n. \tag{4}$$

Stabilization is exactly the event $e_n = 0$.

3 Entry into the bounded-quotient regime

Lemma 3.1 (Entry). *For every $m \geq 1$, there exists*

$$n_0 \leq \lceil \sqrt{2m} \rceil + 2$$

such that $b_{n_0} < n_0^2$. Once $b_n < n^2$ holds, it continues to hold at every later index.

Proof. Let $F(n) = b_n - n^2$. Since $r_n \leq n - 1$,

$$F(n+1) - F(n) = r_n - (2n+1) \leq -(n+2) < 0.$$

Thus F decreases strictly. Starting from $F(2) = m - 4$, summing the displayed decrease gives a negative value within at most $\lceil \sqrt{2m} \rceil$ further steps. Strict decrease gives forward invariance. □

Lemma 3.2 (Bounded quotient and three possible changes). *If $q_n \leq n$, then*

$$\Delta q_n := q_{n+1} - q_n \in \{-1, 0, 1\}, \qquad q_{n+1} \leq n + 1.$$

Consequently, the region $q_n \leq n$ is forward invariant, and every orbit eventually enters it.

Proof. Set $d = 2r_n - q_n$. If $q_n \leq n$, then

$$-(n+1) < d < 2(n+1).$$

The quotient correction in (3) is therefore one of $-1, 0, 1$, and $q_{n+1} \leq q_n + 1 \leq n + 1$.

For the last sentence of the statement, the entry lemma supplies an index with $b_n < n^2$, and then $q_n = \lfloor b_n / n \rfloor < n$; forward invariance of $b_n < n^2$ therefore places every orbit permanently in the region $q_n \leq n$. □

The coordinate (4) satisfies an exact doubling law.

Proposition 3.3 (Exact doubling coordinate). *After entry into $q_n \leq n$,*

$$e_{n+1} = 2e_n - \Delta q_n (n+2), \qquad e_{n+1} \equiv 2e_n \pmod{n+2}. \tag{5}$$

Proof. Since $2r_n - q_n = q_n + 2e_n$, equation (3) gives

$$r_{n+1} = q_n + 2e_n - \Delta q_n (n+1), \qquad q_{n+1} = q_n + \Delta q_n.$$

Subtracting proves (5). □

4 A ratchet for the quotient

The finite-start theorem rests on a simple restriction on decreases of q_n.

Lemma 4.1 (Forced rebound)**.** *If* $\Delta q_n = -1$ *and* $3q_n \le n+1$*, then* $\Delta q_{n+1} = 1$*.*

Proof. A down-step gives

$$q_{n+1} = q_n - 1, \qquad r_{n+1} = 2r_n - q_n + n + 1.$$

The numerator governing the next quotient change is

$$2r_{n+1} - q_{n+1} = 4r_n - 3q_n + 2n + 3.$$

An up-step at the next transition occurs when this is at least $n+2$. That condition is

$$4r_n \ge 3q_n - n - 1,$$

which follows from $r_n \ge 0$ and $3q_n \le n+1$. □

Theorem 4.2 (Ratchet)**.** *Suppose* $3q_k \le k+1$ *for every* $k \in [u, v]$*. Then*

$$q_k \ge q_u - 1 \qquad (u \le k \le v+1),$$

and every decrease of q *in this interval is undone by the next step.*

Proof. Since $3q_k \le k+1$ forces $q_k \le k$, the bounded-quotient lemma applies at every $k \in [u, v]$; in particular $\Delta q_k \ge -1$, so q falls by at most one per step.

We prove by induction on $k \in [u, v+1]$ a statement stronger than the conclusion:

$$\text{either} \quad q_k \ge q_u, \qquad \text{or} \qquad q_k = q_u - 1 \;\text{ and }\; q_{k+1} = q_k + 1. \tag{6}$$

Both alternatives give $q_k \ge q_u - 1$, so the theorem follows.

At $k = u$ the first alternative holds. Assume (6) at some $k \le v$.

If the second alternative holds at k, then $q_{k+1} = q_k + 1 = q_u$, so the first alternative holds at $k+1$.

Otherwise $q_k \ge q_u$. If $q_{k+1} \ge q_u$, the first alternative holds at $k+1$. If instead $q_{k+1} < q_u$, then $q_{k+1} < q_u \le q_k$, so $\Delta q_k = -1$ and $q_{k+1} = q_k - 1$; combined with $q_{k+1} < q_u \le q_k = q_{k+1} + 1$ this forces

$$q_k = q_u, \qquad q_{k+1} = q_u - 1.$$

Because $k \le v$, the forced-rebound lemma applies at k and yields $\Delta q_{k+1} = 1$, that is $q_{k+2} = q_{k+1} + 1$. So the second alternative holds at $k+1$, completing the induction. □

Remark 4.3. The second alternative in (6) cannot be dropped. Tracking only $q_k \ge q_u - 1$ would leave open a flat step at level $q_u - 1$ followed by a further descent; it is the forced rebound attached to each *arrival* at $q_u - 1$ that excludes this, by ensuring the level is never occupied at two consecutive indices. This induction is machine-checked in Lean as `Conjecture.ratchet`, together with the forced-rebound lemma `Conjecture.forced_rebound` and the drop bound `Conjecture.quotient_drop_le_one` that it uses; the axiom audit reports no use of `sorryAx`.

5 The eventual increment bounds the start

Theorem 5.1 (Finite-start bound)**.** *If the orbit from $m \geq 1$ stabilizes with eventual increment c, then*

$$\boxed{m < (c+3)(3c+5).} \tag{7}$$

Proof. Let n_0 be the first index with $b_{n_0} < n_0^2$, and let t be the stabilization index. Then $q_t = r_t = c$. Define

$$S = \{n \in [n_0, t] : 3q_n > n + 1\}.$$

The sequence b_n is nondecreasing, so $m = b_2 \leq b_n$ for every $n \geq 2$.

Suppose first that S is empty. The ratchet theorem on $[n_0, t]$ gives

$$c = q_t \geq q_{n_0} - 1,$$

so $q_{n_0} \leq c + 1$ and

$$b_{n_0} = q_{n_0} n_0 + r_{n_0} < (c+2)n_0.$$

If $n_0 = 2$, then $m < 2(c+2)$, which is stronger than (7). If $n_0 \geq 3$, minimality of n_0 gives $b_{n_0-1} \geq (n_0 - 1)^2$, while monotonicity gives

$$(n_0 - 1)^2 \leq b_{n_0} < (c+2)n_0.$$

This inequality forces $n_0 \leq c + 4$: if $n_0 \geq c + 5$, then

$$(n_0 - 1)^2 - (c+2)n_0 = n_0^2 - (c+4)n_0 + 1 > 0,$$

a contradiction. Hence

$$m \leq b_{n_0} < (c+2)(c+4) < (c+3)(3c+5).$$

Now suppose that S is nonempty, and let $n^* = \max S$. If $n^* < t$, then every $k \in [n^* + 1, t]$ satisfies $3q_k \leq k + 1$. The ratchet theorem gives

$$c = q_t \geq q_{n^*+1} - 1.$$

Since $\Delta q_{n^*} \geq -1$, we also have $q_{n^*+1} \geq q_{n^*} - 1$, and therefore $q_{n^*} \leq c + 2$. If $n^* = t$, the same inequality is immediate from $q_t = c$.

Because $n^* \in S$,

$$n^* + 1 < 3q_{n^*} \leq 3(c+2),$$

so $n^* < 3c + 5$. Finally,

$$m \leq b_{n^*} = q_{n^*} n^* + r_{n^*} < (q_{n^*} + 1)n^* \leq (c+3)n^* < (c+3)(3c+5).$$

□

Corollary 5.2 (Effective finite candidate set)**.** *For each fixed $c \geq 1$, every starting value whose orbit stabilizes with increment c lies in the finite interval*

$$1 \leq m < (c+3)(3c+5).$$

Remark 5.3. The corollary does not assert that every candidate in this interval stabilizes. It says only that no witness for c can occur outside the interval. This distinction is what allows a finite certificate to prove nonattainment without assuming the universal stabilization conjecture.

6 The smallest omitted increments

For $c = 5$ and $c = 7$, Theorem 5.1 gives the complete candidate ranges

$$m < (5 + 3)(3 \cdot 5 + 5) = 160,$$

$$m < (7 + 3)(3 \cdot 7 + 5) = 260.$$

Thus a complete exact computation for $1 \le m < 260$ settles both cases.

The certificate records, for each such m, the first absorption index t, the resulting increment c, and the value $b_t = c(t + 1)$. The rows needed to establish the smaller attained increments are shown below.

Table 1: Explicit witnesses below the first two omissions.

Eventual increment c	Start m	First absorption index t	b_t
1	3	2	3
2	5	7	16
3	13	9	30
4	19	5	24
6	41	7	48

Theorem 6.1 (Certified nonsurjectivity)**.** *The attainable eventual-increment spectrum omits* 5 *and* 7*, while* $1, 2, 3, 4,$ *and* 6 *occur. Hence* 5 *and* 7 *are the two smallest omitted positive integers.*

Proof. The complete certificate contains every start $1 \le m < 260$. No row has eventual increment 5 or 7. By the finite-start bound, this exhausts every possible witness for 5 and every possible witness for 7. Table 1 gives explicit witnesses for $1, 2, 3, 4,$ and 6. □

7 Certificate and reproducibility boundary

The finite part of Theorem 6.1 is deliberately small enough to be reproduced without the larger compressed census used elsewhere in the project. All files named below are in the source repository [4], which also pins the toolchain versions used to produce them.

Certificate

The committed file

`certificates/spectrum_m259.csv`

contains 259 rows, one for each $1 \le m < 260$, with columns

$$(m, t, c, b_t).$$

Its SHA-256 digest is

`66a06cff15735c4a3caf98575f29afbcd881fbef06334616fbc3bc772b7ab084`.

Independent regeneration

The script

`independent/verify_small_spectrum.py`

uses arbitrary-precision integers and the literal recurrence (1). For each start it:

1. iterates until the first index satisfying the absorption criterion;
2. records (m, t, c, b_t);
3. confirms 64 subsequent increments equal c;
4. regenerates the complete CSV and checks its digest;
5. asserts that 5 and 7 are absent and that the witnesses in Table 1 occur.

The reproduction commands are

```
python independent/verify_small_spectrum.py
sha256sum certificates/spectrum_m259.csv
```

On Windows PowerShell, the second command may be replaced by

```
Get-FileHash -Algorithm SHA256 certificates\spectrum_m259.csv
```

Logical boundary

The proof of the finite-start bound is symbolic mathematics. The certificate supplies only the finite exhaustion authorized by that theorem. The nonsurjectivity conclusion does not rely on the project's 10^7-start compressed census, on any heuristic model, or on universal stabilization.

The Lean development in the repository formalizes the foundational identities of Sections 2 and 3, both load-bearing steps of Section 4 — the forced-rebound lemma and the ratchet, including the induction of (6) — and Theorem 5.1 itself, as `Conjecture.finite_start`. The entry lemma is formalized in the form the proof consumes, namely existence of an index with $b_n < n^2$ together with minimality; the explicit bound $n_0 \le \lceil \sqrt{2m}\, \rceil + 2$ is not formalized and is not needed, since the proof of Theorem 5.1 never uses it. The quadratic step $(n_0 - 1)^2 < (c + 2)n_0 \Rightarrow n_0 \le c + 4$ is formalized as `Conjecture.quadratic_step`.

Two boundaries should be stated precisely. First, the formalized theorem takes as hypothesis that the orbit *lies on the absorbing ray* at some index, $b_t = c(t + 1)$ with $c < t$. By the absorption criterion this is equivalent to having eventual increment c, but only the direction ray $\Rightarrow$ constant increment is formalized; the converse uses the limiting divisibility argument in the proof of that criterion and is not. So the machine-checked statement is Theorem 5.1 from the ray hypothesis, and a fully formal route from "the orbit stabilizes with increment c" remains open. Second, Theorem 6.1 is not formalized: its finite exhaustion is checked by an independent arbitrary-precision program rather than inside the proof assistant.

8 Open problems

The principal unresolved question is still Cloitre's conjecture: does every positive start stabilize? The result here separates that question from the spectrum problem. Even if a nonstabilizing orbit exists, Theorem 6.1 remains valid because it concerns the existence of stabilizing witnesses for fixed increments.

Several natural spectrum questions remain:

1. Is the complement $\mathbb{Z}_{>0} \setminus \mathcal{E}$ infinite?
2. What arithmetic structure, if any, governs the omitted increments?
3. Can the quadratic finite-start bound be sharpened substantially?
4. Can nonattainment of a fixed increment be certified by a purely symbolic obstruction rather than finite enumeration?

The larger repository develops structural restrictions on hypothetical nonstabilizing orbits, but those results are intentionally outside the scope of this paper.

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